

Dachuan Song

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Google Scholar | GitHub | LinkedIn | Personal Website

Education

George Mason University , Fairfax, Virginia, USA <i>College of Engineering and Computing</i> Ph.D. in Electrical and Computer Engineering , Current GPA: 3.91 Advisor: Prof. Xuan Wang • Major Courses: <i>Linear Systems, Random Processes, Kalman Filtering, Distributed Control and Optimization, System Identification, Advanced learning from data</i>	Jan 2024 – Present
University of Southampton , Southampton, United Kingdom <i>School of Electronics and Computer Science</i> M.Sc. in Computer Science , GPA: 3.8 (Top 3%) • Major Courses: <i>Advanced Machine Learning, Advanced Deep Learning, Computer Vision, Bayesian, Active & Reinforcement Learning</i>	Sept 2022 – Sept 2023
Xinjiang University , Xinjiang, China <i>School of Software</i> B.Eng. in Software Engineering , GPA: 3.4 (Top 5%)	Sept 2017 – Jun 2021

Publications

CONFERENCE PAPERS AND PREPRINTS:

Elastic Spectral State Space Models for Train-Once Budgeted Inference Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang https://arxiv.org/abs/2601.22488 <i>Accepted to NeurIPS 2026 Main Track (Poster); proceedings forthcoming. arXiv:2601.22488, 2026.</i>	2026
Elastic Selective Spectral Hybrids for Train-Once, Export-Many Budgeted Inference Dachuan Song, Chuchu Chen, Xuan Wang https://arxiv.org/abs/2609.32486	Sep 2026
Mergeable Model-Side Aggregation States for Long-Context Language Models Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang https://arxiv.org/abs/2607.26448	Jul 2026
Causality-based Subject and Task Fingerprints using fMRI Time-series Data Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang https://doi.org/10.1145/3698587.3701342 <i>ACM BCB '24, Article 13, pp. 1–10 (Oral Presentation).</i>	Nov 2024

JOURNAL ARTICLES:

Reconstructing brain causal dynamics for subject and task fingerprints using fMRI time-series data Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang https://doi.org/10.1007/s13755-025-00388-w <i>Health Information Science and Systems 13, Article 70 (2025).</i>	Oct 2025
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Research Projects

Elastic Selective Spectral Hybrids for Budgeted Inference

Jul 2026 – Sep 2026

- Gave each spectral channel a recurrent state with input-dependent decay and write/read gates, so it can retain relevant context and forget distractions. Sliding-window attention handles recent tokens.
- Trained channel count and feed-forward width as capacity tiers, producing compact exports from one checkpoint. At 1.53B parameters, fused batch-one decoding took 1.37 ms/token on a B300, 2.14–2.80× faster than the tested Mamba-2 and Mamba-3 implementations at matched parameter counts.

Elastic Spectral State Space Models for Train-Once Budgeted Inference

Jan 2026 – May 2026

- One training run yields independently deployable models for different budgets. Direct truncation removes ordered Hankel channels and their parameters from each smaller export.
- Input-adaptive gates tune channel contributions; budget dropout trains every retained prefix. Across language, audio, long-range, and offline reinforcement-learning tasks, quality changes smoothly as the budget shrinks.

Mergeable Model-Side Aggregation States for Long-Context Language Models

Jul 2026 – Aug 2026

- Added a 2 KiB mergeable HyperLogLog state beside a frozen language model to track distinct counts and set evidence across context chunks.
- Reached 99.2% accuracy on 3,969 aggregate-then-reason tasks with Gemma 4 (31B), without rereading the full history.

Machine Learning for Brain Activity Patterns using EEG (Master's Project)

Apr 2023 – Oct 2023

- Combined CNN spatial features with LSTM temporal modeling for EEG brain-pattern recognition.
- Used autoencoder pretraining, attention, and transfer learning to extract features from limited EEG data.

Internship Experience

Machine Learning Engineer Intern, Optosurgical

May 2026 – Aug 2026

- Recognized surgical phases and instrument actions in video, providing procedure-state signals to downstream systems.
- Compressed and fine-tuned Depth Anything 3 to 29% of its original size while retaining near-full depth-estimation performance.

Technical Skills

- **Architectures & Inference:** spectral and selective state space models, recurrent memory, sliding-window attention, long-context aggregation, multi-budget model export.
- **ML Engineering:** PyTorch, CUDA, FFT convolution, multi-GPU training, model compression, fused recurrent decoding, reproducible evaluation.
- **Modeling & Control:** linear systems, Kalman filtering, system identification, LQR/MPC, autoregressive models.